

Kunci Jawaban Latihan

KUNCI JAWABAN — SOAL LATIHAN PERTEMUAN 11

STASIUN A: STACK & QUEUE

A1. Stack

Operasi	Stack
push(4)	[4]
push(9)	[4, 9]
push(2)	[4, 9, 2]
pop()	[4, 9] → 2 keluar
push(7)	[4, 9, 7]
pop()	[4, 9] → 7 keluar

b. Data tersisa: **4 dan 9**

A2. Queue

Operasi	Queue
enqueue(8)	[8]
enqueue(3)	[8, 3]
dequeue()	[3] → 8 keluar
enqueue(5)	[3, 5]
enqueue(2)	[3, 5, 2]
dequeue()	[5, 2] → 3 keluar

b. Data tersisa: **5 dan 2**

STASIUN B: SEARCHING

B1. Sequential Search — cari 18

Array: [12, 7, 25, 9, 18, 3, 30, 14]

Langkah	Indeks	Data	Cocok?
1	0	12	×
2	1	7	×
3	2	25	×
4	3	9	×

Langkah	Indeks	Data	Cocok?
5	4	18	

Jumlah perbandingan: 5

B2. Binary Search — cari 15

Array: [3, 7, 12, 15, 20, 25, 30, 35, 42] (indeks 0-8)

Langkah	low	high	mid	Data[mid]	Arah
1	0	8	4	20	15 < 20 → kiri (high=3)
2	0	3	1	7	15 > 7 → kanan (low=2)
3	2	3	2	12	15 > 12 → kanan (low=3)
4	3	3	3	15	Ditemukan!

Jumlah perbandingan: 4

STASIUN C: SORTING

C1. Bubble Sort

Array: [35, 12, 8, 20, 5]

Pas	Array setelah pas
Awal	[35, 12, 8, 20, 5]
Pas 1	35>12→T, 35>8→T, 35>20→T, 35>5→T → [12, 8, 20, 5, 35]
Pas 2	12>8→T, 12>20→F, 20>5→T → [8, 12, 5, 20 , 35]
Pas 3	8>12→F, 12>5→T → [8, 5, 12 , 20, 35]
Pas 4	8>5→T → [5 , 8, 12, 20, 35]

Hasil akhir: [5, 8, 12, 20, 35]

C2. Insertion Sort

Array: [35, 12, 8, 20, 5]

Langkah	Ambil	Array
1	35	[35]
2	12	[12, 35] — 12 < 35, sisip di depan
3	8	[8, 12, 35] — 8 < 12, sisip di depan
4	20	[8, 12, 20, 35] — 20 < 35, sisip di depan 35
5	5	[5, 8, 12, 20, 35] — 5 < 8, sisip di depan

Hasil akhir: [5, 8, 12, 20, 35]

STASIUN D: PSEUDOCODE & FLOWCHART

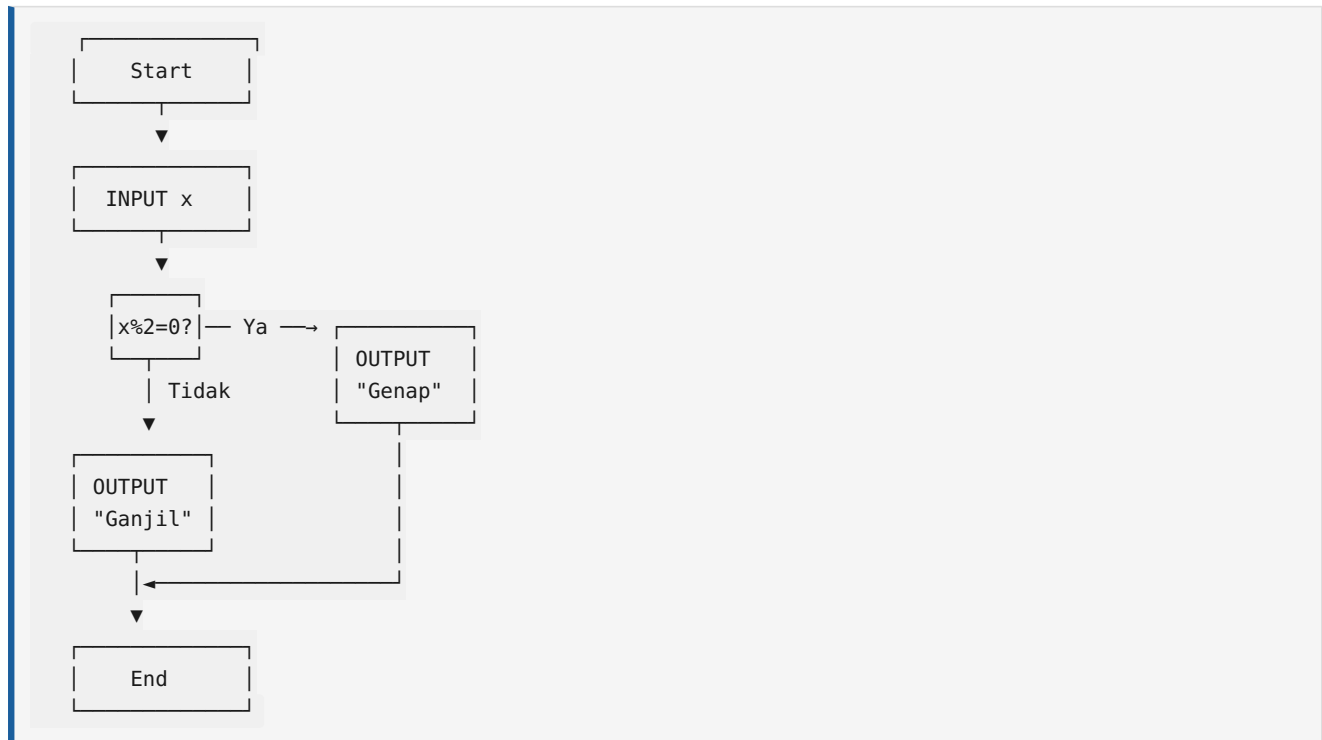
D1. Input 75

$75 \geq 92? \times \rightarrow 75 \geq 83? \times \rightarrow 75 \geq 75?$

$\rightarrow$ OUTPUT "C"

Jawab: C

D2. Flowchart Ganjil/Genap



STASIUN E: PERULANGAN

E1. Output deret genap 1..6

```
FOR i ← 1 TO 6  
  IF i MOD 2 = 0 THEN total ← total + i
```

$i=2 \rightarrow +2, i=4 \rightarrow +4, i=6 \rightarrow +6$ total = $2+4+6 = 12$

E2. Kelipatan 3 dari 3 sampai 30

```
PROGRAM kelipatan_3  
  i ← 3  
  WHILE i <= 30  
    OUTPUT i  
    i ← i + 3  
  ENDWHILE  
END
```

Atau dengan FOR:

```
PROGRAM kelipatan_3
  FOR i ← 3 TO 30 STEP 3
    OUTPUT i
  ENDFOR
END
```

BONUS: Celcius ke Fahrenheit

```
PROGRAM celcius_ke_fahrenheit
  INPUT celcius
  fahrenheit ← celcius × 9/5 + 32
  OUTPUT fahrenheit
END
```

Contoh: $C=100 \rightarrow F = 100 \times 9/5 + 32 = 212$

MGMP Informatika SMAN 6 Cimahi